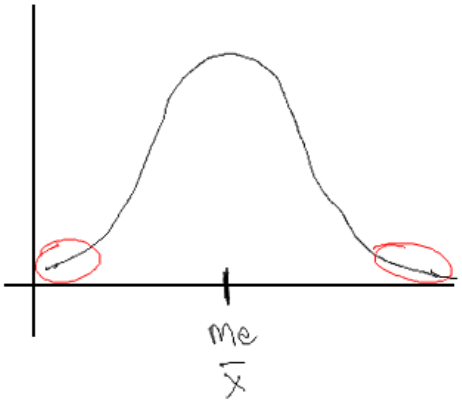
Simetría

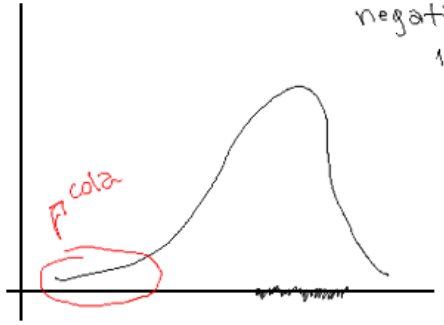


$as = 0$

Asimetría positiva o a la derecha

$as > 0$

Asimetría negativa o a la izquierda



$as < 0$

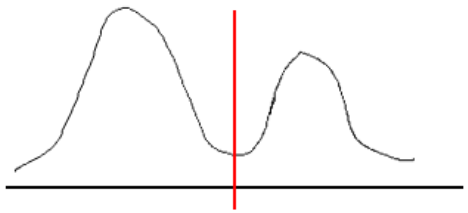
1. $as_1 = 0.43$
2. $as_2 = -0.02$
3. $as_3 = -0.71$ $\longrightarrow$ mas asimétrico

```r
datos |>
   summarise(across(where(is.numeric), ~ skewness(.x)))
```

```
# A tibble: 1 x 4
    Edad Minutos_ejercicio    IMC Presion_sistolica
   <dbl>             <dbl>  <dbl>             <dbl>
1 -0.134            -0.174 -0.318             0.165
```

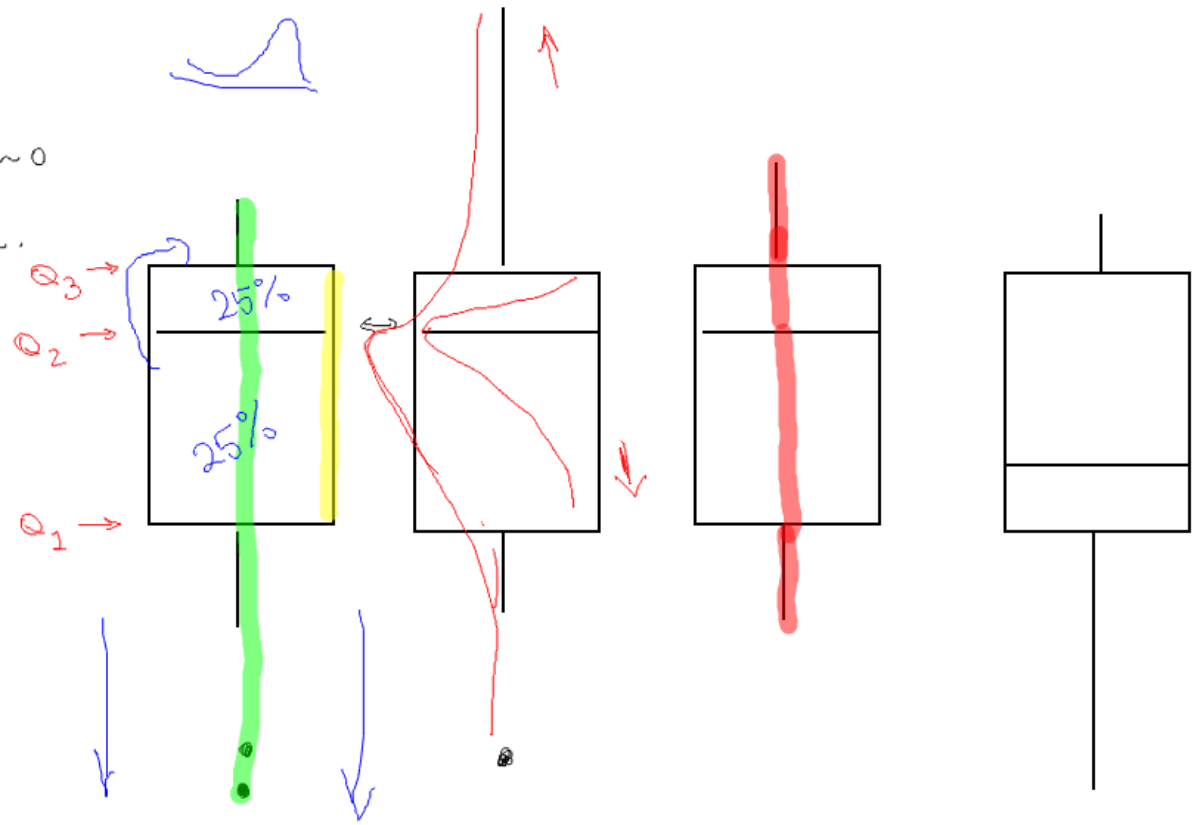
```r
datos |>
   summarise(across(where(is.numeric), ~ bowley(.x)))
```

```
# A tibble: 1 x 4
   Edad Minutos_ejercicio    IMC Presion_sistolica
  <dbl>             <dbl>  <dbl>             <dbl>
1 0.191           -0.0781 0.0196             0.160
```

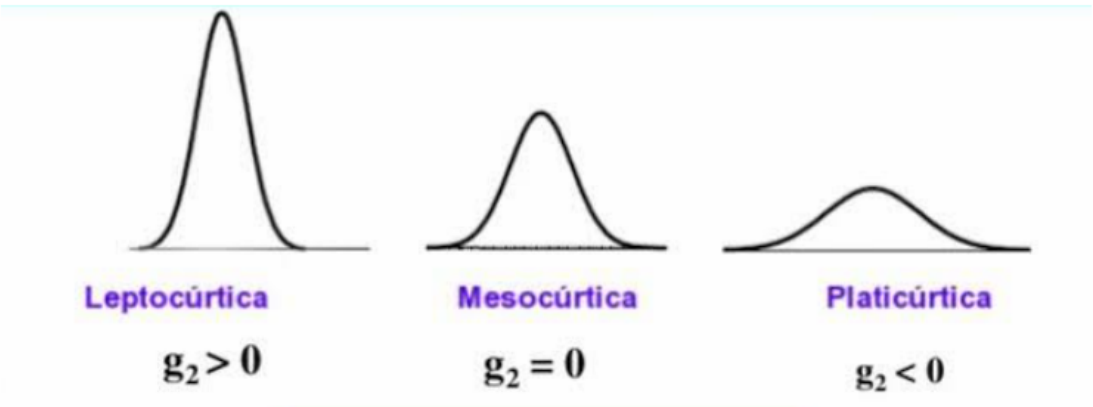


	Asimetría	Asimetría en el 50% central
Edad	Neg.	Pos
Ejercicio	Neg.	Neg
IMC	Neg.	Pos ~ 0
Presión	Posit.	Posit.

	Asimetría	Asimetría en el 50% central
Edad	Neg.	Pos
Ejercicio	Neg.	Neg
IMC	Neg.	Pos ~ 0
Presión	Posit.	Posit.



Curtosis

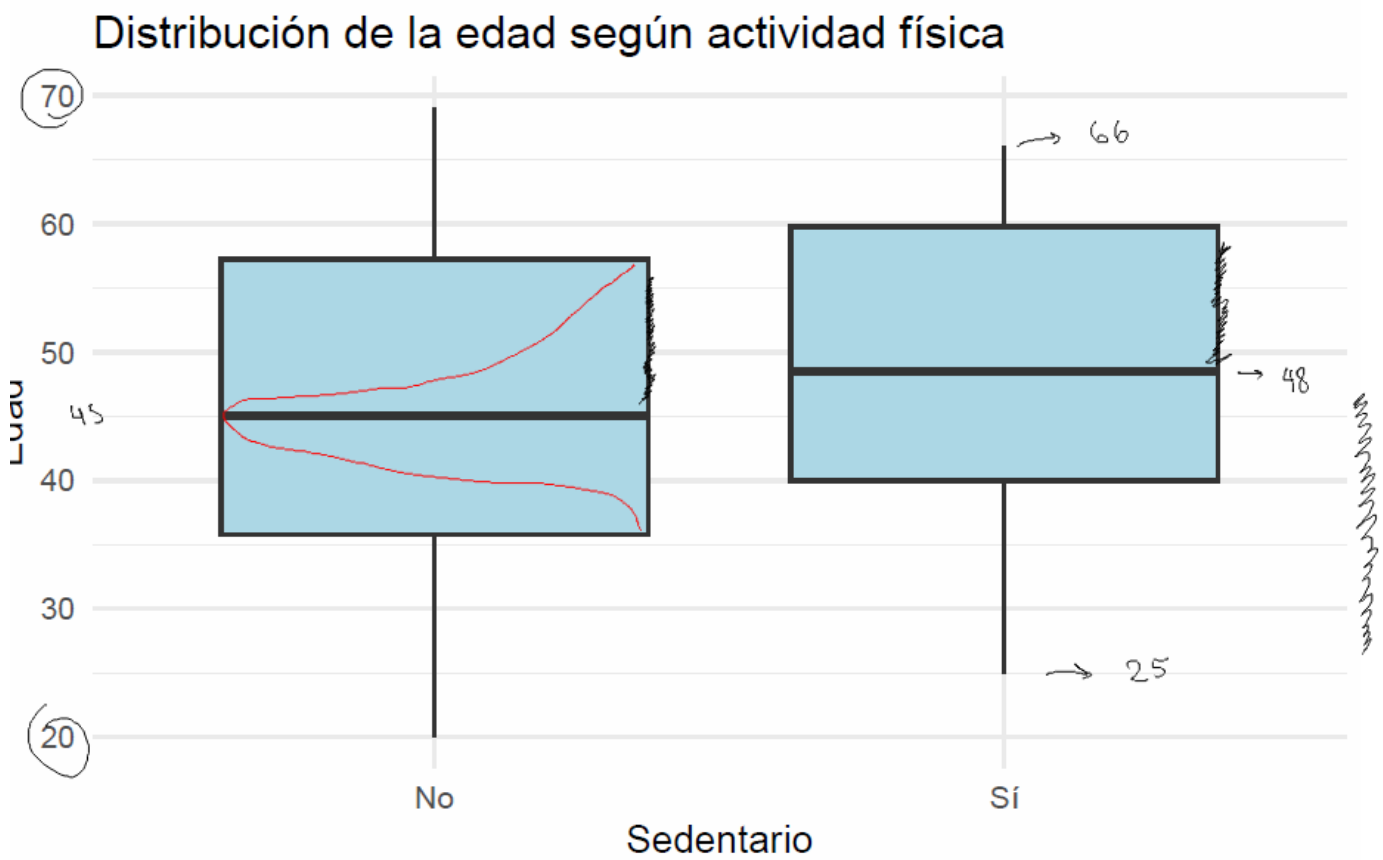


$g_2 > 3$ $\quad$ $g_2 = 3$ $\quad$ $g_2 < 3$

$g_2 \rightarrow 0 \rightarrow$ Mesocúrtica

$g_2 \rightarrow 3 \rightarrow$ Mesocúrtica

$$g_2 = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{s_x^4} - 3$$



100% → Simétrico

50% central → Asimetría positiva

100% → Asimetría negativa

50% central → Asimetría positiva